ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

7wQ9xoyfg1svvu1enrkkgXNa1g7s8xdG2ZPPC9TzBsmv

Date: 5/1/2025
Compounds: Canabis, Cocaine, Fentanyl, Heroin

RESEARCH PURPOSES ONLY
This analysis is generated for research and educational purposes. The predictions and insights provided should not be considered medical advice.

New Compound Analysis

Name: Cannafinol
Formula: $C_{25}H_{30}NO_4$
Weight: 408.52
Drug-likeness: 0.45

Activity:
Analgesic and psychoactive

Target Analysis

Mu-opioid receptor
Binding Affinity: -11
Primary target for opioid analgesics, involved in pain modulation.

CB1 receptor
Binding Affinity: -9.5
Involved in the psychoactive effects associated with cannabis.

Toxicity Profile

Risk Level: High
Affected Systems: Respiratory system, Cardiovascular system, Central nervous system

Mitigation Suggestions:

- Limit dosages
- Continuous monitoring of respiratory function
- Consider opioid antagonists in overdose scenarios

Pharmacokinetics

Absorption:
Rapid absorption through mucosal membranes

Distribution:
Wide distribution, high lipid solubility

Metabolism:
Hepatic via CYP450 enzymes

Excretion:
Renal and biliary excretion

Half-life: 6-8 hours

Detailed Analysis

Cannafinol is a hypothetical compound designed to leverage both opioid and cannabinoid receptors for managing pain and providing mood-altering effects. This novel compound synthesizes key elements of fentanyl, heroin, cannabis, and cocaine, engineered to have selective binding affinity for mu-opioid receptors and CB1 receptors. With its potential for high abuse and addiction, Cannafinol demands rigorous preclinical trials to fine-tune its analgesic benefits against its psychoactive profile. The compound's high lipophilicity ensures rapid central nervous system penetration, aligning with its proposed mechanism of action. While extensive liver metabolism may promise detoxification pathways, the compound's significant toxicity and abuse potential warrant cautious developmental phases.
